LUC@BJTU Weihai Campus

Fundamentals of Communication Systems (CNSCC160) Tutorial 5

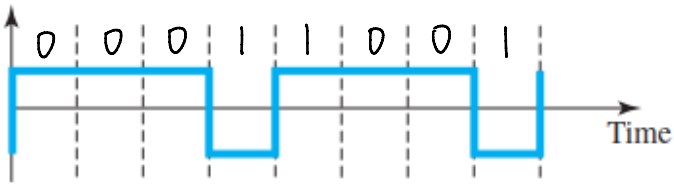
Group Number:

Name:

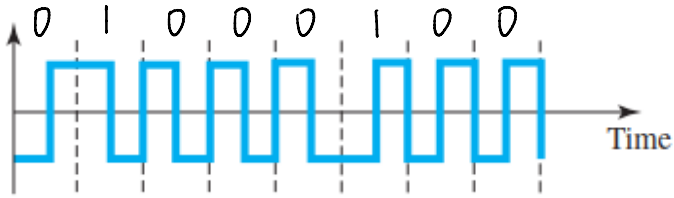
1. Explain the difference between NRZ-L and NRZ-I.

In binary, 0 in NRZ-L represents high voltage, 1 in NRZ-L represents low voltage, 0 in NRZ-I means remain the last voltage. 1 in NRZ-I means reverse voltage.

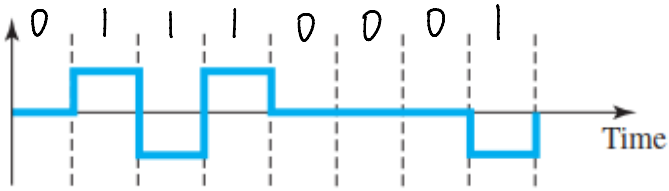
2. Find the 8-bit data stream for each case depicted in the following figure. For differential Manchester, there is no transition at the beginning of the first bit.



a. NRZ-I

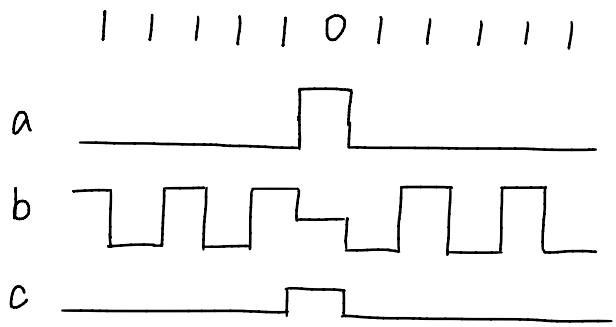


b. differential Manchester



c. AMI

3. Consider a stream of binary data consisting of a long sequence of 1s followed by a zero followed by a long string of 1s, e.g., 11111011111. Assume that the most recent preceding bit 1 for AMI has a negative voltage; and the most recent preceding bit 0 for pseudoternary has a negative voltage. Draw the waveform for this sequence using
   a. NRZ-L
   b. Bipolar-AMI
   c. Pseudoternary



a. NRZ-L has synchronization problem = NRZ-L will create a long stream of 0 and 1 that will cause synchronization.

b. Yes, AMI will create long stream of 0

4. From the waveforms you draw in Q3, think about the following questions:
   a. Does NRZ-L have the synchronization problem? If so, when?
   b. Does AMI have the synchronization problem? If so, when?
   c. Does pseudoternary have the synchronization problem? If so, when?

c. Yes, when pseudoternary creates long stream of 1

5. What are the processes of a PCM encoder?

   Sampling quatization encoding

6. What is the Nyguist sampling rate for each of the following signals? Note that a low-pass signal indicates a signal that occupies the frequency range from 0Hz up to a certain cutoff, while a band-pass signal means a signal containing a band of frequencies not adjacent to zero frequency.
   a. A low-pass signal with bandwidth of 200 KHz? B = 200kHz Sampling rate min = 400kBaud
   b. A band-pass signal with bandwidth of 200 KHz if the lowest frequency is 100 KHz?

   300kHz x 2 = 600kbaud

7. Consider an audio signal with spectral components in the range 300 to 3000 Hz. Assume that a sampling rate of 7000 samples per second will be used to generate a PCM signal.
   a. For SNR = 30 dB, what is the number of uniform quantization levels needed? L
   b. What data rate is required?

SNRdB = 6.02nb + 1.76dB

nb = 4.69

so nb = 5

r = 5, S = 7000

N = 35000